Proposal report

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1 Executive Summary

Currently, tutors are relying primarily on manual processes and human memory to track student progress. This often leads to a lack of reliable and timely signals as to which areas students are struggling with. Similarly, parents often get inconsistent and unclear pictures of how their children are progressing.

Our solution would be to provide tutors with a fully functional pipeline and dashboard that would provide them with all the necessary tools to track student progress with specific skills and ensure that they can get timely signals to detect potentially problematic areas for students.

2 Introduction

The problem that we are tackling in this project is to help teachers track student mastery progress through time. This would allow us to ensure parents and students get timely feedback in a clear and concise manner. Knowledge tracing is a machine learning method to track and predict a student's mastery over time based on their performance data.

We have 3 main objectives for this project :

- Create a mastery tracker that uses knowledge tracing to estimate student's mastery probability
- Create an early warning system that flags areas that students are struggling with
- Create a dashboard application visualises the elements above and other students insights (e.g average scores)

The outcome of the project would be a fully functional and reproducible pipeline that updates the dashboard when given new data.

3 Data & Exploratory Data Analysis (EDA)

The dataset comes from an AP Statistics course, with data collected over 7 months. It contains information about the course structure (how assignments are structured), the knowledge structure (how KC relationships are structured) and the student's performance (student scores) for 25 students. The dataset spans across 33 assignments (includes homeworks and exams for a total of 895 items) and 256 knowledge components, for a total of 21,808 item-level student records.

The observational unit is the student item interaction, which represents a single student's response to a specific assignment question. The key variables are the student's scores, the knowledge component IDs (indicate which KCs each item assesses) and the KC graphs (provide information about how the KCs are related to each other).

3.1 Key insights

3.1.1 Missing values

Some students did not hand in assignments which has led to missing values in the data.

Table 1: Number of students that are missing assignments.

Number of missing assignments	Number of students
0	8
1	12
2	3
4	1
5	1

Looking at Table 1, there are more students that are missing at least one assignment (17 students) than students not missing any (8 students). However, most of the students with missing assignments are only missing one (71% of students with missing data), which implies that the missingness is relatively evenly spread out across students.

Table 2: Number of assignments that are missing students.

Number of missing students	Number of assignments
0	14
1	11
2	8

Looking at Table 2, the number of assignments that are missing students (58% of assignments) is close to the number of assignments that aren't missing students (42% of assignments). This means that the missing data is relatively evenly spread across assignments, with no assignment missing more than 2 out of 25 students (8%). It also suggests that the missingness isn't due to an assignment specific issue.

The missing data seems to be sparse and well-distributed across the observations, with no assignment missing more than 2 students and most students (71% of students with missing data) missing only 1 assignment. Overall, this suggests that missingess is unlikely to introduce significant bias into the analysis.

3.1.2 KC coverage imbalance

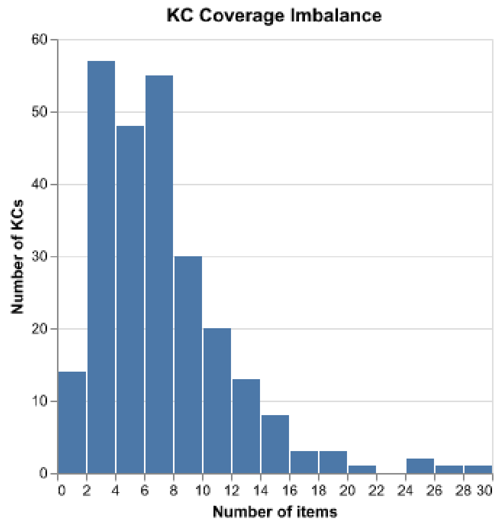

Figure 1: This figure shows the distribution of KC coverage across assignment items. Most KCs are covered by fewer than 10 items, with coverage peaking at 2-3 items per KC. A small number of KCs are covered by up to 30 items.

Looking at Figure 1, the KC coverage is heavily right-skewed, with a median of 6 items per KC. This imbalance means that some knowledge components contain up to 725 observations, while others are not assessed at all (4 KCs with 0 observations). The difference in coverage among KCs could affect the reliability of mastery estimates for under-covered KCs.

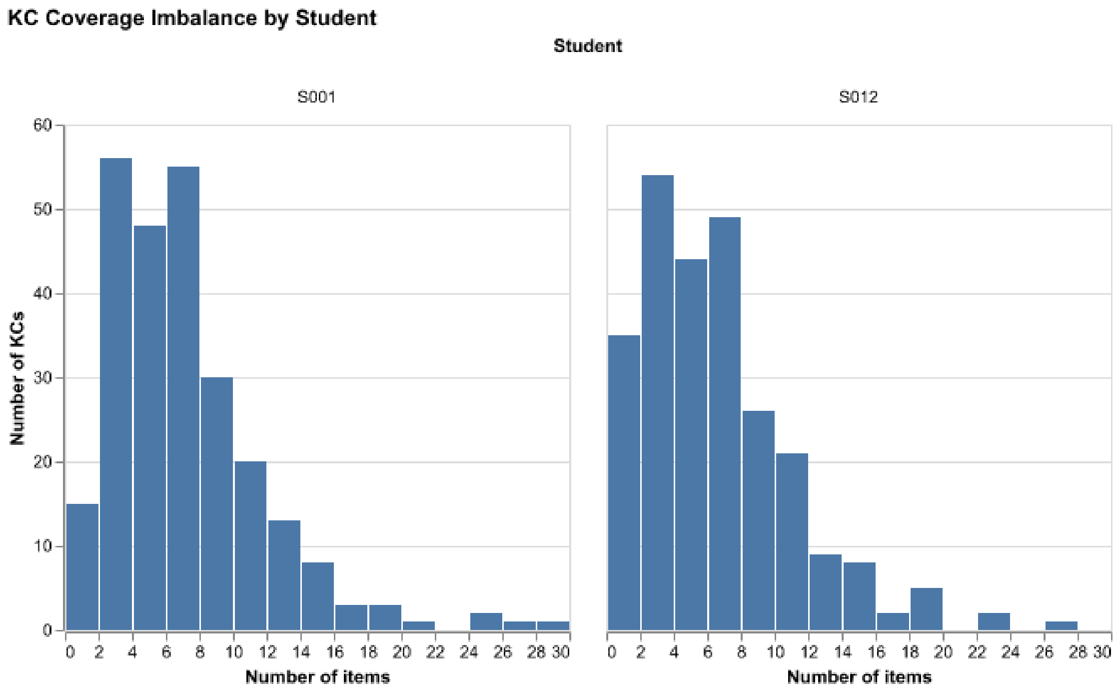

Figure 2: This figure shows the distribution of KC coverage across assignment items for 2 different students, one who handed in all assignments (left : S001) and one who has 5 missing assignments (right : S012).

Looking at Figure 2, both students have a similarly right-skewed KC coverage distribution, though S012's is shifted slightly more towards the left compared to S001, which is consistent with the 5 missing assignments. In fact, S012 has 18 untested KCs compared to S001 who only has 4, meaning that already under covered KCs are even sparser for S012. Overall, this means that having missing assignments may reduce the reliability of mastery estimates.

3.1.3 Performance band imbalance

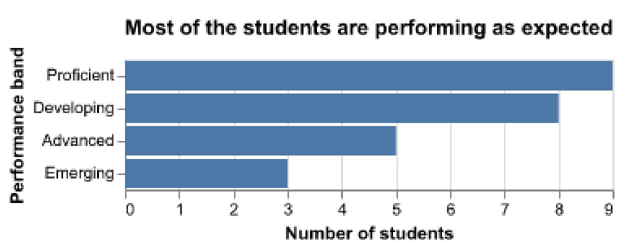

Figure 3: This figure shows the distribution of students across four performance bands, ordered from least to most proficient: Emerging, Developing, Proficient, and Advanced. The majority of students fall in the Proficient (9) and Developing (8) categories, indicating that most are performing at or near the expected level.

Looking at Figure 3, there is an imbalance in the performance band distribution, with there being 3 times as many proficient students (9) as emerging (3). This might lead to some prediction issues in the model building phase, since classifiers tend to provide less accurate estimates for underrepresented groups. This can be especially problematic since emerging students are precisely those that the early warning system should reliably flag.

3.1.4 Knowledge component relationships

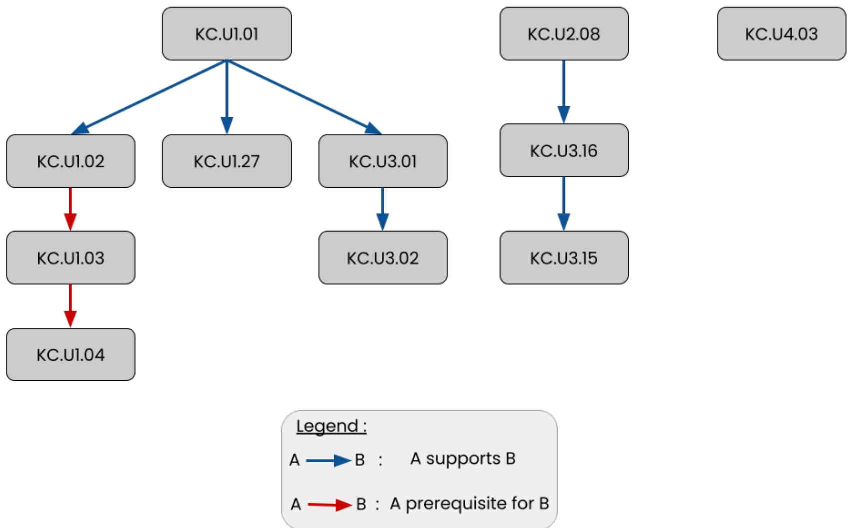

Figure 4: This figure shows the prerequisite structure for a subset of KCs. Arrows indicate the type of relationship (support or prerequisite) between KCs.

Some knowledge components have prerequisite or support relationships, which means that students need to master the foundational KCs before progressing to the dependent ones. As shown in Figure 4, these dependencies can span multiple levels (e.g KC.U1.02 is a prerequisite for KC.U1.03 which itself is a prerequisite for KC.U1.04). This hierarchical structure needs to be taken into account in the modelling phase, since a student's performance might depend on how well they've mastered the prerequisites.

4 Data Science Approach

Our approach uses knowledge tracing methods to estimate the probability that a student has mastered a knowledge component, based on their assignment scores and the KC relationship structure. These mastery estimates will be used in a dashboard which will contain insights into student performance along with 3 main components : a knowledge profile that summarises each student's strengths and gaps, an early warning system that flags students that are struggling, and personalised suggested next steps.

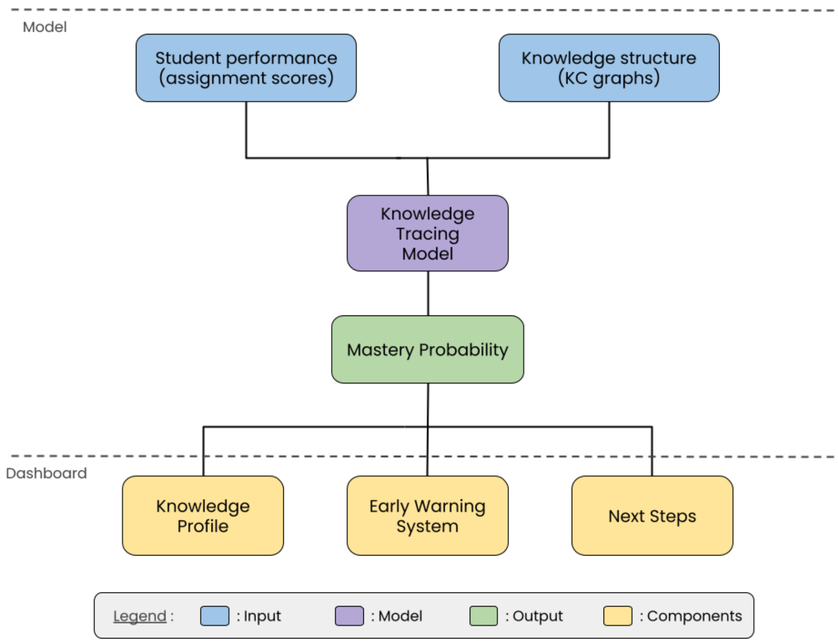

Figure 5: This figure is an overview of the proposed pipeline. From student performance data and knowledge structure inputs, through knowledge tracing, to three dashboard components : knowledge profile, early warning system and next steps.

4.1 Model

To estimate student mastery we will use Knowledge Tracing, which is a family of models that track the progression of a student's knowledge as they interact with knowledge components (through assignments). Since the dataset contains item level student responses related to specific KCs, this method is a natural fit as it is designed to use this structure to estimate unkown mastery states. Understanding a student's knowledge state can help teachers customize the lesson plans to better suit individual needs, as well as help students identify the skills they are struggling with the most.

We will build and compare multiple different types of Knowledge Tracing models to determine the one that is the best fit for our data. Our baseline model is a Bayesian Knowledge Tracing (BKT) model, which assumes that it is possible to estimate a student's latent knowledge state (their understanding) based on their observed performance (their grades) (Shen et al. (2024)). We chose this model as our baseline since it is simple and easily interpretable, which makes it a natural choice to which compare more complex models to. However, the significant KC coverage imbalance (identified in the EDA section above) means that simpler models like BKT that treat all KCs as independent, may struggle with sparse KC coverage. More complex models that share information across KCs may therefore perform better on this dataset.

These models will produce an estimate of the probability that a student has mastered a specific KC based on their past performance (assignment scores), which is called the mastery probability. These estimates are the foundation for all three dashboard components : the knowledge profile, the early warning system and the suggested next steps.

4.2 Dashboard

A major problem teachers face when explaining a student's progress is the lack of visual tools to show their evolution over time. Dashboards can help bridge that gap by providing an overview of student performance, allowing teachers to understand how their students are progressing in a single glance (Aleven et al. (2010)). Our approach is to build two dashboard views that will allow teachers to gain both class and individual level insights.

4.2.1 Dashboard Components

4.2.1.1 Knowledge profile

Using the mastery probability obtained from the Knowledge Tracing model, we will build a knowledge profile for each student. Each profile will display all KCs and their corresponding mastery probability which will allow both students and teachers to get a clear overview of a student's knowledge state. These profiles will be updated each time new assignment data is entered, which will help ensure that they reflect the student's most recent performance.

4.2.1.2 Early Warning System

For every student, the Early Warning System will automatically flag knowledge components that they appear to be struggling with, based on these two conditions:

- The mastery probability remains below a certain threshold (e.g. 50%) after a predefined number of interactions (e.g. 3 assignments)
- The mastery probability has become stagnant and shows no meaningful increase across consecutive interactions

These conditions allow the system to flag both students who are struggling to grasp a KC and those whose understanding is not progressing. Flagged KCs will be highlighted on the dashboard to ensure timely interventions from teachers. The thresholds for flagging a students will be determined emperically by comparing student performance progression across multiple assignments.

4.2.1.3 Next Steps Suggestion System

A student is considered to have mastered a KC if their mastery probability is above a certain threshold (e.g. 95%). Once a student has mastered a KC, the Next Step Suggestion System will recommend assignments that cover KCs that they have not mastered yet, prioritising KCs whose prerequisites have been satisfied. This helps ensure that recommendations follow a logical progression through course content.

4.2.2 Prototypes

To help ensure the teachers can get both individual and class level insights, we will be building two dashboard views : a student view and a teacher view.

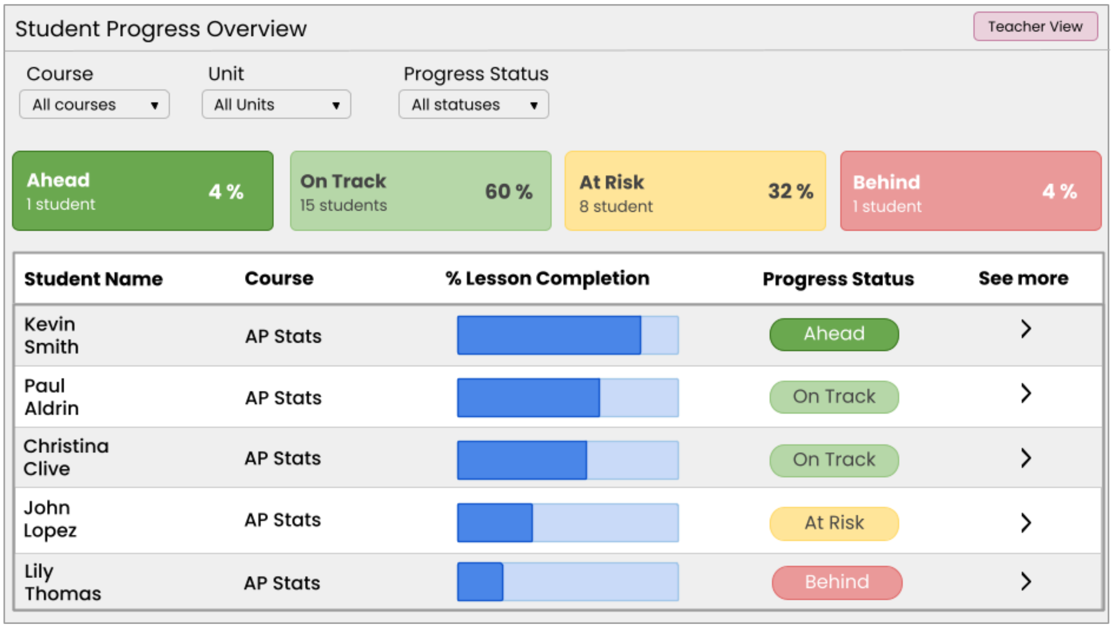

Figure 6: This is a prototype of the teacher view dashboard, showing class-level progress. Adapted from Jasen (2020)

The teacher view provides an overview of the entire class's performance. As shown in Figure 6, it displays each student's lesson completion progress and categorises students into four progress bands : Ahead, On Track, At Risk and Behind. This will help teachers to quickly identify students who may need additional help.

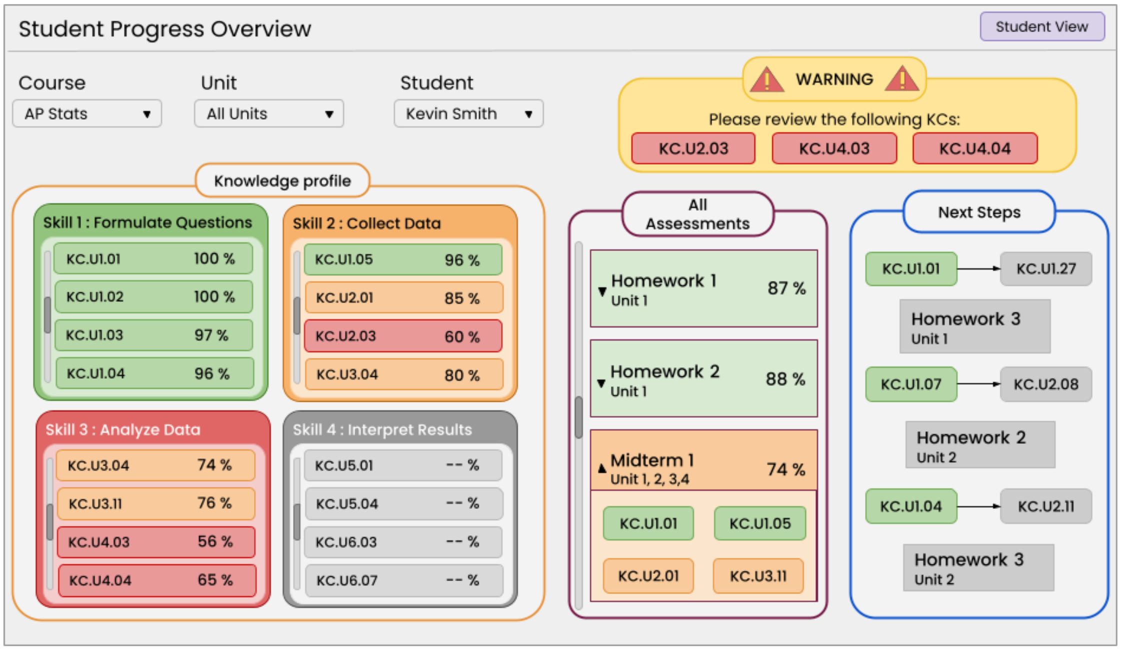

Figure 7: This is a prototype of the student view dashboard, showing individual student progress through a knowledge profile, early warning panel and next steps suggestion panel.

The student view of the dashboard, shown in Figure 7, gives a detailed breakdown of how an individual student is progressing. It includes three main components : a knowledge profile (summarises mastery across skill areas), an early warning panel (flags KCs that require attention) and a next step suggestion panel (suggests the next assignments a student should attempt).

5 Evaluation & Success Criteria

5.1 Evaluation

We will be evaluating our approach on three fronts, as follows :

5.1.1 Model Validity

To preserve the dependency structure between KCs, the data will be split into train and test sets by student. We will be evaluating the model performance by comparing the mastery probability to the final exam scores of the students. Using RMSE to determine how far the predicted mastery probabilities are from the actual test scores (smaller the RMSE the stronger the predictive validity of the model).

5.1.2 Model Comparison

To find the model that is the best fit for our data, we will be comparing any model we fit to the baseline BKT model. Using RMSE and AIC/BIC we will be comparing the model fit with model complexity to ensure the final model fits all requirements.

5.1.3 Early Warning System Validity

We will create a dataset using the student's performance band from Overall_Scores where Emerging students are labelled as Flag and the others are labelled as Not Flag. Using this dataset, we will be evaluating the Early Warning System as a binary classification problem with metrics like AUC, Precision and Recall to ensure that only necessary warnings are sent.

5.2 Success Criteria

We will consider our project a success if all the following conditions are met :

- The mastery tracker reliably estimates student mastery
- The early warning system reliably signals student struggle
- The dashboard provides a clear and reliable way to show student progress
- The dashboard provides teachers with class-wide descriptive statistics inclusive of skills, knowledge components and overall status of a class
- The dashboard provides students with personal progress, areas of concerns alongside "what to practice next"

6 Timeline

The project will be split into weekly milestones. Each milestone contains a list of elements which must be completed by the given deadline.

Milestone	Due Date	Description
Milestone 1	May 8	- Pipeline that outputs model ready dataset - Bayesian Knowledge tracing model (model build and analysis)
Milestone 2	May 15	- Alternative models (model build and analysis) - Dashboard skeleton design
Milestone 3	May 22	- Final dashboard design - Final model selection
Milestone 4	May 29	- Dashboard update pipeline - Early warning system
Milestone 5	June 25	- Production ready product

References

Aleven, Vincent, Franceska Xhakaj, Kenneth Holstein, and Bruce M McLaren. 2010. "Developing a Teacher Dashboard for Use with Intelligent Tutoring Systems." *Technology* 34: 44–50.

Jasen. 2020. *Student Progress Tracking Dashboard*. Flight Schedule Pro. https://www.flightschedulepro.com/blog/student-progress-tracking-dashboard.

Shen, Shuanghong, Qi Liu, Zhenya Huang, et al. 2024. "A Survey of Knowledge Tracing: Models, Variants, and Applications." *IEEE Transactions on Learning Technologies* 17: 1858–79.